

A Theoretical Model for Teacher Knowledge Development in AI-Supported Data Science Instruction

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Abstract

While the importance of teacher knowledge for data-focused and AI-supported instruction is recognized, less is known about how that knowledge develops as teachers move from learning to designing to implementing. To address this, the current study conducted a comparative case study examining how two in-service science teachers develop knowledge for AI-mediated data analysis across three interconnected professional learning phases: Teacher-as-Learner (TAL), Teacher-as-Designer (TAD), and Teacher-as-Implementer (TAI). Using the TPACK framework and Schön's reflective practice theory as conceptual lenses, the study traces how reflection, knowledge development, and pedagogical decision-making unfold as teachers participated in the professional learning. Importantly, this paper does not provide the empirical results of the study but rather the theoretical model used for understanding how teachers knowledge can be developed in data-focused and AI-supported instruction. Implications of this theoretical model are also provided.

Keywords

teacher knowledge, TPACK, data analysis, large language models, AI in education, professional development, reflective practice, K-12 science education, comparative case study

1 Introduction and Background

1.1 Data Practices in Science Learning

The development of data skills is important to support students' learning of science, and there have been calls to engage students in such data practices [1, 7, 8, 16–18, 22, 24]. However, students tend not to have opportunities to engage in such practices because teachers often lack the necessary skills and knowledge of data practices and the tools [8, 15], which are themselves skills students need [6]. Furthermore, although using first-hand and second-hand data can be beneficial for learning [9], second-hand data are often messy and characteristic of real-world data, creating a need for computational tools and practices for working with them [8, 16].

Despite the availability of data and tools such as TinkerPlots, CODAP, MyWorld GIS, and Microsoft Excel and machine learning techniques that can support data work [2, 4, 7, 23, 27], teachers experience a number of difficulties, some of which are: lack of confidence leading data-focused discussions [13]; reliance on familiar tools and relatively small datasets that may provide fewer opportunities to develop practices needed for working with real world

data [24], and perceived technological inadequacy and challenges integrating computational tools into instruction [23]. These challenges point to a broader problem, which is that teachers need to develop their knowledge and be supported in data-focused science instruction. However, little is known about how that knowledge develops.

1.2 LLM-Based Tools and Teacher Knowledge for AI-Mediated Work

These challenges suggest a potential role for Large Language Model (LLM)-based tools in lowering some technical barriers to data-intensive practices. Unlike conventional data science tools, which often require learning specialized interfaces and procedures, LLM-based tools can provide computational capabilities via natural-language interactions with a minimal learning curve [5, 28]. Despite their computational affordances, LLM-based tools do not remove the need for teachers to develop knowledge for integrating AI, data, science content, pedagogy, and classroom context. In this study, teachers' knowledge is conceptualized through the Technological Pedagogical Content Knowledge (TPACK) framework [20, 21] (Table 1).

I also consider how each of these knowledge domains interacts with the others based on the TPACK framework.

1.3 Professional Learning as a Developmental Trajectory

Engaging in and integrating data work into their classroom practice may require teachers to shift, adapt, and reconsider their knowledge of data, science, AI tools, and pedagogical moves during classroom practice. Such knowledge development is unlikely to be captured by examining what teachers know at a single point in time. Hence, to examine the trajectory of this knowledge development, I develop a professional development (PD) across three interconnected phases: Teacher-as-Learner (TAL), Teacher-as-Designer (TAD), and Teacher-as-Implementer (TAI). These three phases, as I will later describe in the methods section, are analytically useful because TAL makes teachers' emerging knowledge and initial ways of reasoning visible; TAD shows how that knowledge is transformed into instructional plans; and TAI provides an opportunity to examine how those plans are enacted, challenged, and adapted in classroom contexts. As such, they provide a developmental arc through which the study traces how teachers' knowledge is experienced, mobilized, enacted, evolved, and potentially reconstructed.

Table 1: TPACK Domains as Conceptual Lenses for This Study

TPACK Domain	Description
Technological	LLM-based tools capabilities and limitations, prompting and interaction strategies, data access and manipulation, output evaluation, and appropriate use of tools
Pedagogical	Structuring and decision-making in AI-supported data-focused instruction
Content	Subject-matter expertise and data science practices

Reflective practice provides a lens for examining this knowledge development process. Schön’s theory distinguishes reflection-in-action, in which practitioners reconsider their knowing while engaged in practice, from reflection-on-action, in which they retrospectively examine practice [26]. Building on this work, reflection-for-action describes reflection directed toward subsequent practice [12]. These forms of reflection are embedded in all activities across TAL, TAD, and TAI and provide a way to examine how teachers interpret their experiences, reconsider their knowledge, anticipate instructional needs, and make or revise pedagogical decisions.

Reflective practice, for example, allows us in TAL to establish a baseline of teachers’ knowledge, TAD to make their intended instructional decisions visible before implementation, and TAI to examine how those intentions are enacted, challenged, and adapted in the realities of classroom practice.

While the importance of teacher knowledge for data-intensive and AI-supported instruction is recognized, less is known about how that knowledge develops as teachers move from learning, to designing to implementing. Hence, I followed two science teachers as they moved from learning about AI and data practices to designing and implementing AI-supported, data-focused instruction. This study, thus, provides a comparative, phase-oriented account of that developmental process and asks three questions:

- **RQ1 (Process):** How do teachers engage in reflective practice (reflection-in-action, reflection-on-action, and reflection-for-action) as they learn about, design with, and implement AI tools for data science instruction?
- **RQ2 (Substance):** What shifts occur in teachers’ perceptions, subject matter knowledge, and pedagogical content knowledge for AI-mediated data science across the three phases of the professional learning experience?
- **RQ3 (Experiential):** What aspects/elements of the AI-data science professional learning experiences were influential in shaping their reflective practice, knowledge development, and pedagogical decisions when integrating AI into data-focused instruction?

2 Methods

This study employs a comparative case study design [14, 29], with each of the two in-service science teachers treated as a single case. This study method allowed treating each teacher as a case and enabled an in-depth examination of how reflective practice, knowledge development, and pedagogical decision-making unfolded within each teacher’s trajectory and in relation to the specific experiences and contexts of that case. Comparing the cases then

made it possible to examine similarities and differences in how these processes developed across teachers. Hence, I first constructed a within-case account of each teacher’s trajectory across TAL, TAD, and TAI and then conducted a cross-case comparison to identify similarities and differences in these developmental trajectories.

2.1 Study Design and Context

The study context was a professional development program aimed at in-service STEM teachers using AI to support data practice. The program adapts a three-phase framework for teacher professional development spanning 4 hours each, for a total of 12 hours [11, 19]. The three phases were designed as interconnected opportunities for teachers to move from learning and exploring, to designing, to implementing and adapting AI-supported data instruction.

First, during the Teacher-as-Learner (TAL) phase, teachers learned data skills and used AI to support their data practices. The activities involved teachers learning about and participating in hands-on sessions about Large Language Models and data practices as described in the Data Investigation Process framework of Lee et al. [15]. The teachers directly engage with LLMs and data practices, allowing them to develop and interrogate emerging understandings of what the tools can do, how data can be worked with, and how these practices might relate to science instruction. I argue that teachers cannot meaningfully design and enact AI-supported data instruction without first developing some understanding of LLM capabilities, data science processes, and how these may be useful in their classrooms. This learning experience was designed by the researchers to model what the teachers may enact with their students.

Second, in the Teacher-as-Designer (TAD) phase, teachers co-designed instructional materials and tools with the facilitator, taking into account their students, the contextual situation of the learning environment (e.g., access to tools), and what they learned during the TAL phase. The activities involved, amongst others, identifying relevant datasets, developing AI prompt templates for students, and redesigning their existing lesson units. This phase is more pedagogically oriented than TAL and focuses on transforming their developing knowledge of data practices and AI to inform pedagogical decisions and preparation. Our position is that knowing how to use an LLM-based tool for data work is different from knowing how to structure an activity so that students can investigate a dataset with an LLM in alignment with instructional goals.

Third, Teacher-as-Implementer (TAI) involved teachers enacting their designed teaching and learning materials and tools in the classrooms and reflecting on their successes, challenges, and potential iterations. This phase allows understanding of the difference

between what teachers intended to do in TAD, what actually happened in classroom implementation, and how teachers responded to those realities. It tests teachers' knowledge in context, where students' prior knowledge, time constraints, technology access, unexpected tool behavior, and curriculum demands may require teachers to adapt in real-time. This phase was characterized by check-ins, facilitator visits to the teachers' classrooms, a post-TAI reflection interview, and responding to a questionnaire.

This design allowed the study to trace how reflection was connected to changes in knowledge and pedagogical decision-making as teachers moved across TAL, TAD, and TAI (Table 2; Figure 1). Importantly, throughout the phases, I used second-hand and large datasets [9, 24], supported teachers in navigating the difficulties of finding and using such datasets [25], and provided the necessary support [15].

2.2 Participants

This study used purposive sampling to recruit two experienced in-service high school science teachers by word-of-mouth to participate in a professional development opportunity. Both of them taught science subjects at high school level, they had different levels of experience using AI, and participated in the PD associated with this study individually during different academic terms. Based on participants' characteristics, the study materials differed slightly in volume but not conceptually. Both participants consented to participate in the study in accordance with the Institutional Review Board (IRB).

2.3 Data Collection

I collected data across all three phases and distinguished primary, secondary, and contextual data. The primary data are (1) audio-visual Zoom recordings of the TAL and TAD sessions (totaling eight hours), which provide teachers' interactions with LLM-based tools, their data practices and their reflection think-alouds that capture their moment-to-moment reasoning and decision-making; (2) records of TAI classroom enactment (totaling about four hours) in the form of post-implementation questionnaire, classroom observation field notes and phone call check-ins—all of which capture their enacted practice, real-time decisions, contextual constraints, and their reported interactions among students and tools; and (3) reflection interviews of approximately 1 hour conducted after each phase, which reveal how teachers made decisions and their self-reported evolving knowledge.

Secondary data are teacher artifacts, each serving a distinct analytic role. The teachers provided their lesson units, which showed us their intended instructional design, including the planned roles of AI and students, the data practices involved, and pedagogical scaffolds. The teachers also shared their LLM chat logs, which revealed their prompting, tool use, and their appraisal and adaptation of LLM outputs. They also completed written reflections as the activities in TAL and TAD progressed, showing us their retrospective science, perceived learning, design rationale, and anticipated changes. Finally, I conducted a 1-hour pre-study interview, which served as contextual and baseline data. It established each teacher's prior experience and informed the design of ecologically valid and relevant activities throughout the PD.

2.4 Data Analysis

Our analysis followed a sequential, phase-oriented comparative case approach where each analytic step informed the next. First, I used interaction analysis [10] on the TAL and TAD recordings to examine teacher-LLM activity, including how teachers operated and adapted tools, appraised LLM outputs, investigated data, clarified questions, and surfaced constraints during their data-focused learning and design activities. The interactional episodes provided evidence of teachers' emerging knowledge, reflection, and decision-making during engagement with AI-supported data practices.

Second, I applied thematic analysis [3] to the reflection interviews. Here, I identified reflection-on-action, where teachers retrospectively examined a completed activity, and reflection-for-action, in which they articulated intentions for subsequent practice. I then traced forward into later phases to determine whether and how they were enacted.

Third, I analyzed design artifacts and chat logs to establish teachers' intended designs and their appraisal of LLM outputs. Across these steps, I coded knowledge using the TPACK domains and their intersections, and used reflective practice to understand how this knowledge evolved. All through, I compared data across sources to trace chains linking reflection, knowledge development, and pedagogical decisions.

These analyses were organized at three levels. I first analyzed both teachers' data in turn within TAL, TAD, and TAI (e.g., Teacher 1: TAL followed by Teacher 2: TAL; Teacher 1: TAD followed by Teacher 2: TAD). In doing that, I noted each teacher's developmental trajectory across the three phases, examining what changed, what persisted, what was transformed or abandoned, and what the teacher carried forward. I then compared the two trajectories to identify convergences and divergences, and examined how differences in teachers' prior experiences, contextual conditions, and PD components may account for these differences. Table 3 maps each research question to the phases, data sources, and analyses used to answer it.

2.5 Trustworthiness

I addressed trustworthiness primarily through triangulation across data sources and phases, comparing what teachers did in interaction data, what they designed in artifacts, and what they reported in interviews and written reflections. Analytic memos and an audit trail of coding and interpretive decisions will be maintained to document how evidence was moved from phase-level observations to within-case and cross-case claims.

3 Expected Contributions and Implications

The study design described demonstrates how a phase-oriented comparative case approach can be used to examine the development of teachers' knowledge for AI-mediated data practices across professional learning and classroom implementation. The three-phase study design enables tracing how TPACK is developed, mobilized, and adapted across a developmental arc from learning to enactment.

The study design has implications for four areas. For AI-focused teacher professional learning, the three-phase structure offers a

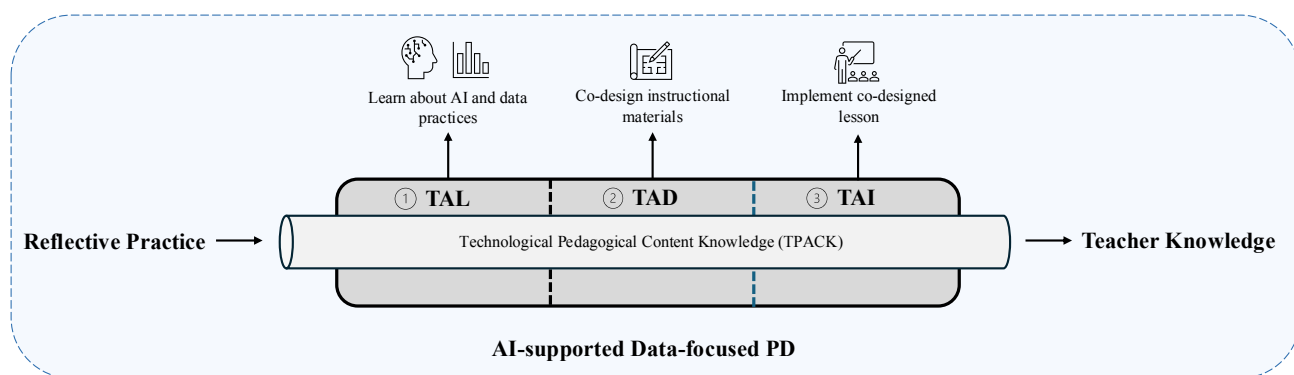


Figure 1: Conceptual framework of the AI-supported data-focused professional development program. Teachers move through three interconnected phases: Teacher-as-Learner (TAL), Teacher-as-Designer (TAD), and Teacher-as-Implementer (TAI); representing a developmental arc from learning about AI and data science, to co-designing instructional materials, to implementing co-designed lessons in classroom contexts. Reflective practice is integrated in all activities across the three phases to understand how teacher knowledge evolved, while TPACK serves as the developmental medium through which teacher knowledge is built, mobilized, and adapted. The study traces how this trajectory shapes the development and enactment of teacher knowledge for AI-mediated data science instruction.

Table 2: Phase-Oriented Design: Roles, Knowledge, Reflection, and Analytic Purpose

	TAL	TAD	TAI
Teacher role	Learner	Designer	Implementer
TPACK	Initial technological, data-science, content, and pedagogical knowledge through experiencing AI-supported data science	Transform emerging knowledge into instructional designs responsive to students, curriculum, context, and AI affordances/limitations	Enact, evaluate, and adapt those learning and designs under authentic classroom conditions
Reflective practice	Reflecting on emerging experiences and for design	Reflecting on design decisions and for implementation	Reflecting in, on, and for classroom practice
What this phase makes possible	Gives teachers knowledge and experiences to draw upon when designing	Makes teachers' pedagogical reasoning and intended decisions visible before classroom implementation	Reveals how knowledge and design hold up, change, or break down in practice

model for designing PD that sequences learning, design, and implementation to make knowledge development visible and traceable. What I present here also extends the TPACK framework to AI-mediated data work, addressing an underexplored yet increasingly relevant topic for K–12 science teachers. Furthermore, I describe how reflection-in-action, reflection-on-action, and reflection-for-action connect to knowledge development and pedagogical decision-making across a sustained professional learning experience. Finally, the study design provides insight into the kinds of knowledge and reflective practices that support teachers in integrating LLM-based tools into data-focused instruction.

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Table 3: Alignment of Research Questions, Phases, Data Sources, and Analyses

RQ	TAL	TAD	TAI	Primary data	Analysis
RQ1 Reflective practice	Reflection on learning and for design	Reflection on design and for implementation	Reflection in, on, and for action	Recordings, think-alouds, reflection interviews, written reflections	Interaction analysis, thematic analysis, forward tracing of reflection-for-action
RQ2 Perceptions and TPACK	Emerging TPACK	Mobilized and reconfigured TPACK	Enacted and adapted TPACK	Recordings, interviews, artifacts, chat logs	TPACK coding, cross-phase comparison, intended versus enacted decisions
RQ3 Influential elements	Hands-on exploration, facilitator modeling, and encountering LLM errors	Co-design, dataset selection, prompt templates, contextual constraints	Student responses, unexpected AI behavior, check-ins, adaptation	Interviews, observations, teacher artifacts, written reflections	Thematic analysis, triangulation, and cross-case comparison of conditions

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